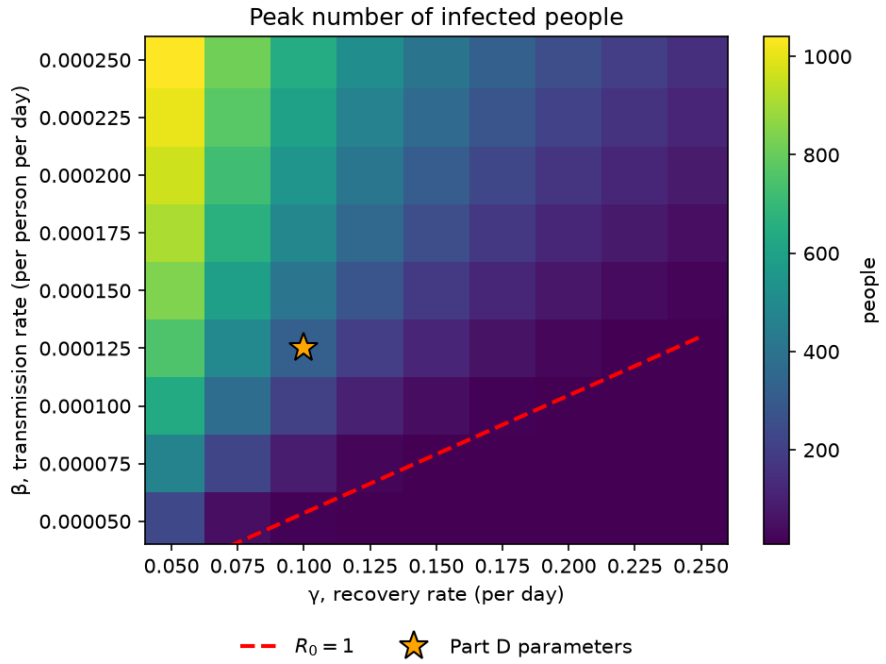
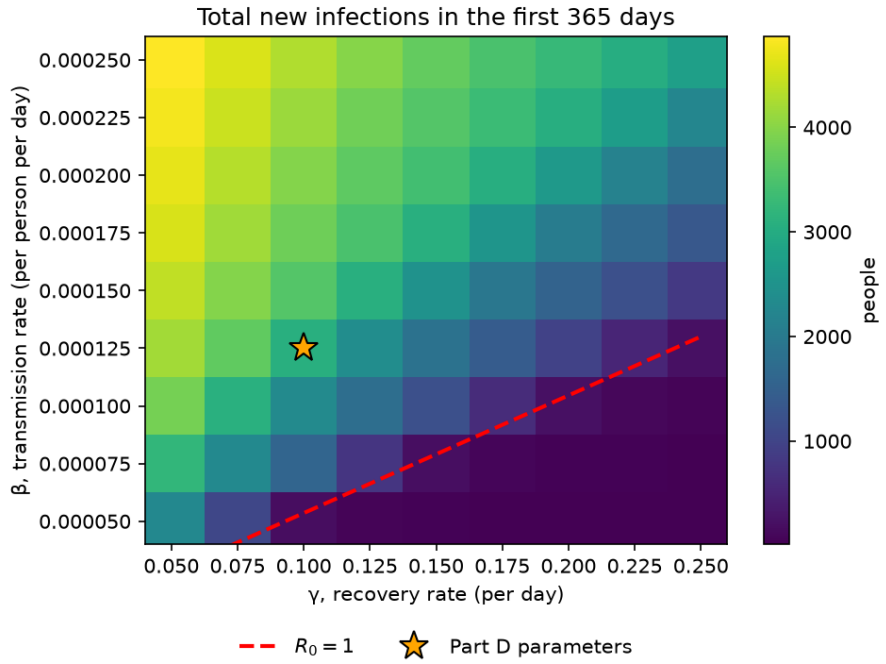
# SIR and SEIR Model Implementation for Pandemic Spread

**Hypothesis:** The maximum infected population will peak at day 40 in the SIR model, while the SEIR model will show attenuated waves with added births, and beta will have the dominant affect on the infected peak.

**Key Number:** Over the same factor of 5 range, gamma moved the peak from 9 to 753 infected and beta from 13 to 644.




**Conclusion**: SIR model peak size was overestimated (470 vs 600 to 700) while the peak day was close (43 vs 40), SEIR model displayed expected wave form with unpredicted non-zero baseline, and gamma affects infection peaks more than expected.

**Modeling lesson:** visualizing the effect model parameters is not comprehensive and testing and plotting on real or synthetic data elucidates true behaviors to inform future modeling.